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RESEARCH ARTICLE

Ensemble Machine Learning Model for Inner Speech Recognition: A Subject-Specific Investigation

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ABSTRACT Inner speech recognition has gained enormous interest in recent years due to its applications in rehabilitation, developing assistive technology, and cognitive assessment. However, since language and speech productions are a complex process, for which identifying speech components has remained a challenging task. Different approaches were taken previously to reach this goal, but new approaches remain to be explored. Also, a subject-oriented analysis is necessary to understand the underlying brain dynamics during inner speech production, which can bring novel methods to neurological research. A publicly available dataset, “Thinking Out Loud Dataset,” has been used to develop a Machine Learning (ML)-based technique to classify inner speech using 128-channel surface Electroencephalography (EEG) signals. The dataset is collected on a Spanish cohort of ten subjects while uttering four words (“Arriba,” “Abajo,” “Derecha,” and “Izquierda”) by each participant. Statistical methods were employed to detect and remove motion artifacts from the signals. A large number (191 per channel) of time-, frequency- and time-frequency-domain features were extracted. Eight feature selection algorithms are explored, and the best feature selection technique is selected for subsequent evaluations. The performance of six ML algorithms is evaluated, and an ensemble model is proposed. Deep Learning models are also explored, and the results are compared with the classical ML approach. The proposed ensemble model, by stacking the five best logistic regression models, generated a promising result with an overall accuracy of 81.13% and an F1 score of 81.12% in the classification of four inner speech words using surface EEG signals.

INDEX TERMS Inner speech, brain-computer interface (BCI), machine learning, classification, convolutional neural network (CNN), deep learning, logistic regression, ensemble model.

I. INTRODUCTION

Language and speech processing has always been an intriguing and complicated topic in neurological research. This is because a number of cerebral structures and networks

associated with factors like emotional processing, contextual meaning, and cognitive regulation collectively add to the core language system [1]. A number of models have been proposed that underline the necessity of dorsal connection to explain the neuroanatomy of language in a simplified manner [2], while other investigations have provided evidence showcasing the intricate composition of the arcuate

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fasciculus alongside the superior longitudinal fasciculus [3], [4], [5]. This neural pathway establishes connections between numerous target regions within the frontal cortex and multiple target regions within the temporal and parietal cortex. Moreover, investigations utilizing functional imaging techniques to study healthy subjects predominantly reveal bilateral activations and the involvement of a large network that exhibits partial activity depending on specific language tasks [6]. This finding contrasts with classical models, such as the Broca or Wernicke models, which primarily emphasize the study of the left hemisphere. Interestingly, motor system and sensorimotor pathways have also been found to be involved in language and speech processing [7], [8]. Different speech components such as phonetic properties, semantic content, and even abstract linguistic constructs related to the articulated or inner speech appear to be grounded in the motor system [8], [9], [10]. This connection gives rise to numerous interactions between the motor system and the complex processes of language perception and speech production.

Electrophysiological signals recorded from the brain have been an integral part of research on brain dynamics for decades. Such research has helped scientists understand how the brain processes language and various speech forms. Electrocorticography (ECoG) or intracranial Electroencephalography (iEEG), Stereo Electroencephalography (SEEG), and Electroencephalography (EEG) record brain activity; however, surface EEG has long been the signal of choice due to its non-invasive nature and practicality. Typically, EEG signals are acquired by placing electrodes on the scalp and capturing them with EEG recording systems [11]. EEG signals are valuable inputs for a variety of applications, including emotion detection [12], rehabilitation [13], and, most importantly, speech recognition [14]. Numerous studies have been conducted to classify brain activities associated with various components of language and speech production, including syllables, phonemes, vocals, and words. Zhao and Rudzicz [15] studied seven phonetic and syllabic cues during covert speech production and categorized them. In more recent works, Jahangiri et al. [16], [17] analyzed four phonemic structures. Other studies, including Cooney et al. [18], Tamm et al. [19], and Ghane and Hossain [20], analyzed EEG signals generated during the imagined speech production of five vowels. However, studies focusing on minute components such as vowels, phonemes, and syllables are often limited to a narrow scope of applicability, necessitating additional research on a broader aspect of language and speech. In this regard, there have been other studies that were carried out involving imagined words. González-Castañeda et al. [14] were among the first to classify imagined words (“up,” “down,” “left,” “right,” and “select”), yet their results were evaluated only subjectively. Pawar and Dhage [21] extended this line of research using extreme learning machines but focused on generalized performance across subjects rather than subject-specific neural dynamics. Nguyen et al. [22] employed Riemannian manifold features, grouping imagined words by length, though their study

was constrained by small sample sizes and limited channel selection. More recent studies attempted CNN-based decoding (e.g., Cooney et al., 2019 [18]), but their performance was constrained by data scarcity and convergence issues. In this work, channel selection was done based on the investigations into 2-3 subjects out of 15. Thus, it is evident that a subject-specific investigation into EEG signals to decode language and speech perception and production has not been explored enough in the literature. This calls for more research using such an approach to decode brain signals for classifying inner speech. The lack of publicly available EEG dataset to classify words from inner speech makes this a challenging task.

Recent developments in EEG-based inner speech recognition highlight the integration of CNNs, RNNs, and hybrid ensemble approaches to address data scarcity and inter-subject variability [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25]. These works underscore the limitations of deep models when applied to small subject-wise datasets and motivate feature-driven ensemble methods such as the one proposed in this study.

In contrast, our work introduces a subject-specific ensemble framework, optimized with MRMR-based feature selection, to decode four inner speech words. A dataset prepared by recording EEG signals during inner speech production in Spanish language has been utilized for this matter. Motion artifacts have been removed before extracting features from the raw data. Then, feature-selection has been carried out and inner speech is classified using Machine Learning (ML) techniques for each subject. Unlike earlier studies that often overlooked inter-subject variability, we explicitly evaluate subject-wise topographies, demonstrating that individual brain dynamics substantially influence decoding accuracy. This methodological novelty enables our approach to outperform previous EEG-based inner speech classification attempts on the same dataset. The main contributions of this study are as follows:

- Motion artifacts were detected using statistical methods and were removed by discarding the low-frequency components using variational mode decomposition (VMD) on the channels exhibiting such artifacts as a pre-processing step.
- 2D visualization techniques such as topographical maps were utilized to identify differences between different states like resting, visual processing, and inner speech production. This was also employed to understand the differences among the subjects in producing inner speech as well as to observe the trends for different words.
- Several time-, frequency-, and time-frequency-domain features were extracted from the EEG signals, followed by utilizing various feature selection algorithms to enhance the performance of ML models.
- Several classical ML-based techniques, 1D signal and 2D spectrogram-based deep learning (DL) techniques were evaluated using the processed as well as the raw

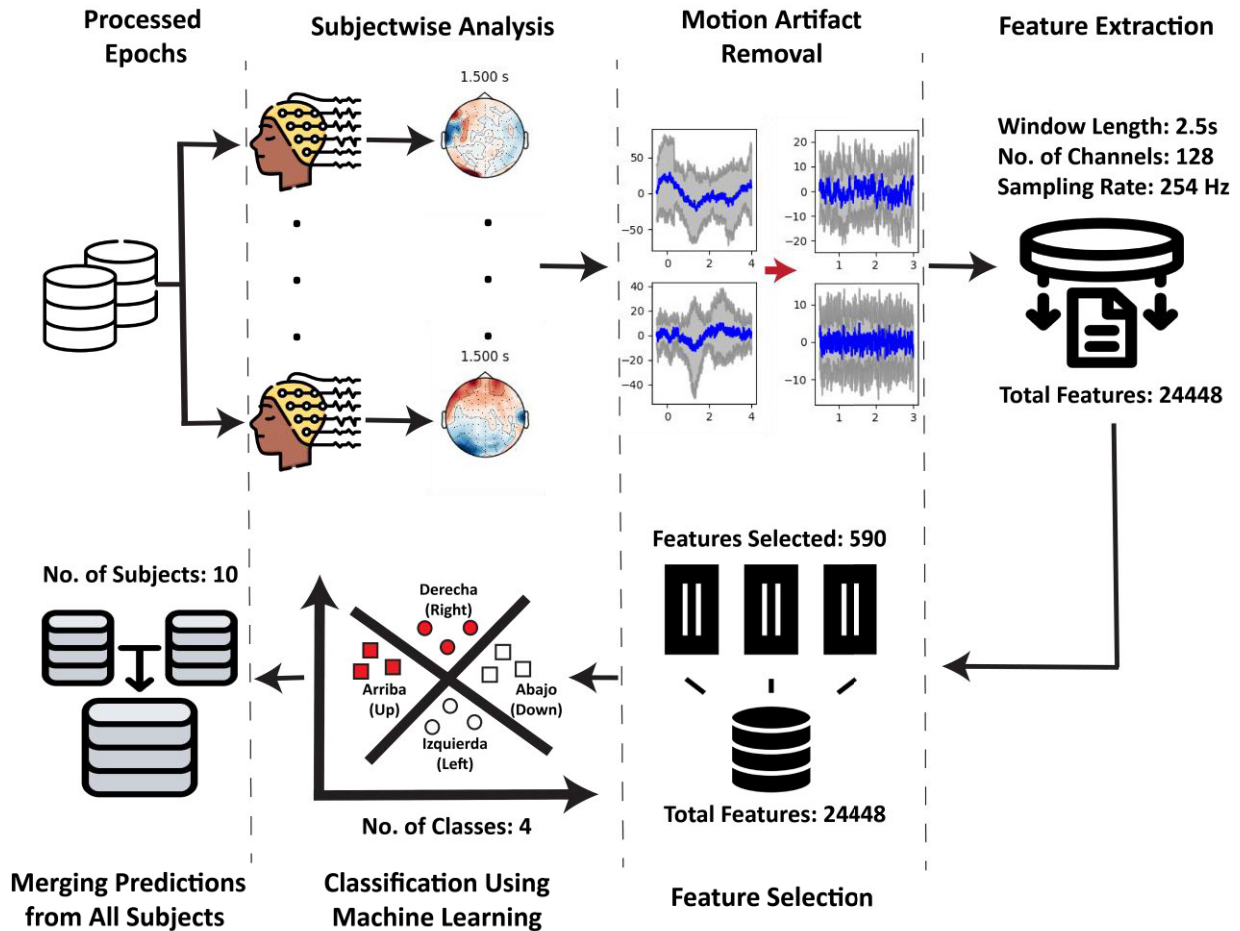


FIGURE 1. Proposed framework for inner speech recognition. The pipeline includes: (1) raw EEG acquisition, (2) preprocessing and artifact removal, (3) feature extraction in time, frequency, and time-frequency domains, (4) feature selection (MRMR), (5) classification using ML/DL models, and (6) ensemble model stacking for final prediction.

data, followed by proposing an ensemble model to classify the inner speech.

- A publicly available dataset for inner speech has been utilized to evaluate the proposed methodology, and the results obtained from our approach surpassed the existing works on this dataset by a significant margin.

The novelty of this work is twofold: (i) a subject-specific ensemble model that leverages MRMR-selected EEG features to capture individualized neural dynamics, and (ii) an integrated pipeline of VMD-based artifact cleaning, feature engineering across time-frequency domains, and ensemble stacking. Unlike previous generalized models [14], [21], [22], we systematically analyze subject-wise differences and demonstrate that tailoring models per subject yields superior accuracy (81.13%). Furthermore, we propose future exploration of domain adaptation and Transformer-based deep architectures, which may enable cross-subject generalization while preserving individualized representations.

Section III outlines the results and discussion obtained from all the methods described in the previous section

through quantitative and qualitative analysis. Section IV presents brief conclusions from this study.

II. MATERIALS AND METHODS

For the identification of inner speech from EEG signals, a publicly available dataset [23] has been used. This section provides a detailed discussion of the study materials and methods, which contains dataset description, data processing steps taken by the authors of the dataset, Event-Related Potential (ERP) analysis of the EEG data, motion artifact removal process, feature extraction steps, feature selection methods, and the experiments using different ML and DL models. The overall proposed framework is shown in Figure 1.

A. “THINKING OUT LOUD DATASET” DESCRIPTION

A publicly available dataset shared by Nieto et al. [23] has been used in this study to evaluate the proposed approach. The dataset was collected from a Spanish cohort while the ten participating subjects were uttering four words (“Arriba,” “Abajo,” “Derecha,” and “Izquierda”). These correspond to

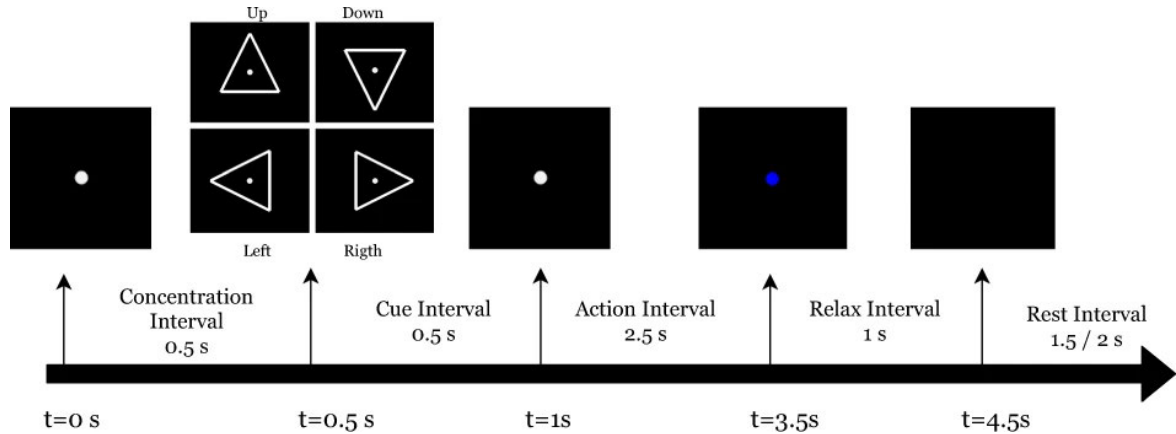


FIGURE 2. The data collection setup. In each trial, one of the four cues (“Up,” “Down,” “Right,” “Left”) were shown to the participant [23].

“Up,” “Down,” “Right,” and “Left.” The dataset includes two types of speech activities: (i) pronounced speech and (ii) inner speech, also referred to as imagined speech, recorded using 128-channel EEG along with EOG and EMG signals recorded with a 24 bits resolution and a sampling rate of 1024 Hz using “ActiveTwo system, BioSemi” amplifier [23]. The external channels facilitated motor and eye movement detection during inner speech activities to tag erroneous experiments, as well as investigations for pronounced speech detection. Each trial by the participants lasted for 4.5 seconds, of which the corresponding speech activity was performed for 2.5 seconds. A detailed outlook of a sample trial is shown in Figure 2. The data collection process involves concentration, cue, action, relaxation, and rest interval. The data collection was conducted in 3 sessions, in which the subjects completed 200 trials in each of the first 2 sessions. However, based on the fatigue and willingness of the subjects to carry out further trials, the number of experiments in the third session varied for different subjects.

B. DATA PROCESSING

Some processing of the data was done by the authors of this dataset [23]. At first, the data were re-referenced using two external channels as the data collection was done using a reference-free acquisition system. Then, a zero-phase band-pass filter with finite impulse response was applied to the data, the lower and upper bounds being 0.5 Hz and 100 Hz, respectively, followed by applying a notch filter of 50 Hz. After that, the data were decimated to obtain a sampling rate of 256 Hz before segmenting the data. Then, Independent Component Analysis (ICA) was employed on the EEG channels to capture sources, and it aided in detecting unwanted motions, which were discarded during the reconstruction of EEG signals to generate the final dataset. Also, EMG signals were used to detect mouth movements during inner and visualized condition actions, which were used for tagging the trials containing such movements. The raw data, as well as the processed data, are provided in the publicly available

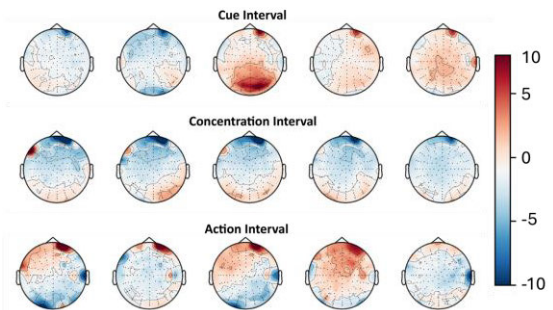


FIGURE 3. Topographical maps of cue interval, concentration interval, and action interval for subject 10.

dataset [23]; however, the processed data is used for this study.

C. TOPOGRAPHY ANALYSIS OF EEG DATA

Event Related Potentials (ERPs) are measured using 128 channels EEG amplifier and are used to investigate the behaviour of different brain regions. ERPs are time-locked to specific events or stimuli, such as visual, auditory, or cognitive stimuli. They represent the brain’s electrical activity in response to these events and provide insights into the neural processes underlying perception, attention, memory, and other cognitive functions [24]. From ERPs, topographical maps can be constructed, which can provide insights into the activities of different regions during any state.

Analyzing the topographies of different subjects and for different intervals in this dataset reveals that the activities of different regions of the brain were different for all intervals, with more activity in the visual cortex during the cue interval (when participants were shown visual information about the four cues) and intense activities in specific locations during the concentration and action interval, when participants were engaged in focused attention and action, as evident in Figure 3. In addition, it is observed that the ERPs of distinct words uttered internally by a particular subject shared some

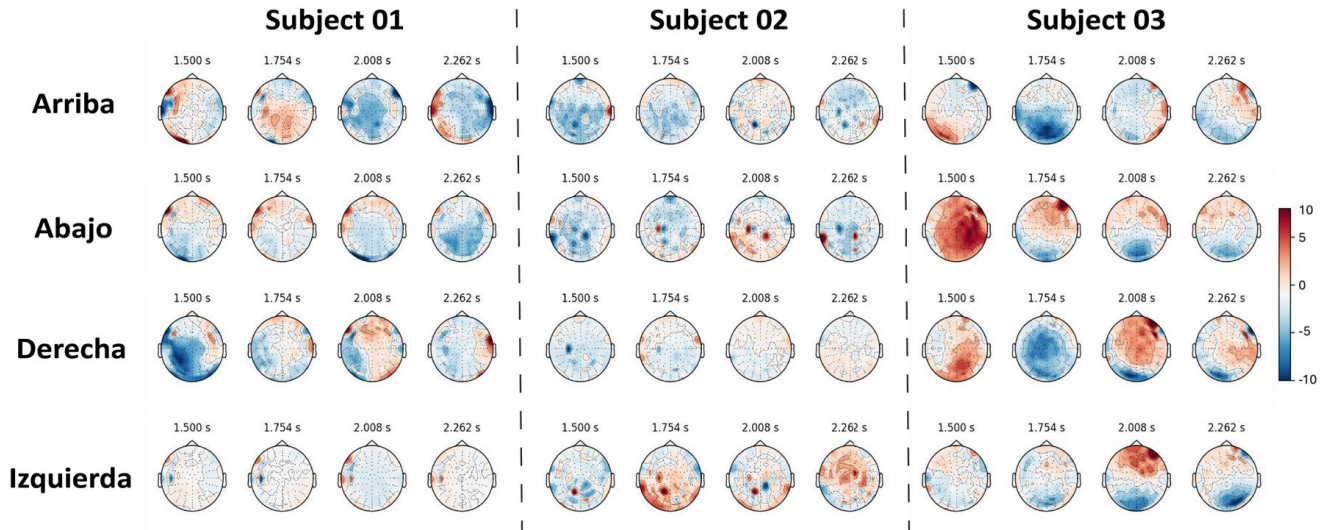


FIGURE 4. Topographical maps of three subjects with all four words uttered during inner speech tasks.

similarities, although these similarities were weak for the majority of subjects. However, there is no similarity among the topographies of different subjects. Figure 4 suggests that inner speech production of each subject was unique. This necessitates subject-specific data investigation and evaluation of various approaches.

D. MOTION ARTIFACT REMOVAL

Even though some measures were taken to eliminate artifacts from the raw data [23] by the dataset provider, the processed data may still contain motion artifacts. The presence of motion artifacts can substantially hinder the detection of inner speech; therefore, these artifacts must be eliminated for the accurate identification of words from EEG signals. The channels that exhibit such artifacts are identified in this study by closely examining the signals of all the channels of various subjects using mean and standard deviation plots as shown in Figure 5. These artifacts have to be removed in order to clear up the data. Methods such as Discrete Wavelet Transform (DWT) [25], Wavelet Packet Decomposition (WPD) [26], Empirical Mode Decomposition (EMD) [27], Ensemble Empirical Mode Decomposition (EEMD) [28], Singular Spectrum Analysis (SSA) [29], and Variational Mode Decomposition (VMD) [30] were explored previously to remove motion artifacts [31], [32]. VMD is utilized in this study to remove motion artifacts due to its resistance to sensitivity and noise, as well as its non-recursive method for estimating modes concurrently [32]. Using VMD, six modes are generated from the data of each trial, and the Intrinsic Mode Function (IMF) corresponding to the mode with the lowest frequency is discarded [33]. Figure 5 depicts the entire motion artifact removal procedure. After that, the clean data is used to reconstruct topographical maps, and it is observed that ERP drifts have been removed (Figure 6 & supplementary Figure S1).

E. FEATURE EXTRACTION

Time-domain (TD) [34], frequency domain (FD) [35], [36], and time-frequency domain (TFD) [37] features have been extracted from the EEG signals of the action interval of each subject. The TD features consist of pure TD characteristics and other statistical features (such as mean, deviation, absolute value, etc.) obtained from the signal in the time domain [38]. Features obtained from the envelope of the signal have also been included under TD features. Features that describe the shape and size of the signal in the frequency domain representation have come under the FD features. These features consist of both simple features, such as dominant frequency, and more complex features, such as spectral roll-off point and spectral deformation. Some other FD features have also been calculated, such as skewness and kurtosis of the power spectrum. The TFD features in this study are mainly obtained using discrete wavelet transform (DWT) [39], [40]. The features consist of calculating the band power, mean absolute value, waveform length, root mean square value, standard deviation, and fractal length of the extracted wavelets. In total, 50 TD, 24 FD, and 117 TFD features have been extracted from each EEG channel segment in this study. The total number of features per channel was 191. The details of all the features from different domains are provided in the Supplementary Tables T1, T2, and T3. The dataset contains data from 128 EEG channels. As a result, a total of 24448 features (191×128) were extracted from the data of each subject. Such a huge number of features can cause a high variance in the prediction using ML models [41] and hence requires feature selection techniques.

F. FEATURE SELECTION TECHNIQUES

Following feature extraction, selecting features has become crucial since the huge number of features is likely to cause a significant downgrade of model performance due to

overfitting. With that in mind, the following feature selection techniques are explored:

1. ANOVA F [42]
2. Chi-Square [43]
3. Mutual Information (MI) [44]
4. Pearson-Coefficient (PC) [45]
5. Decision Tree-Based Selection [46]
6. Principal Component Analysis (PCA) [47]
7. ReliefF [48]
8. Minimum Redundancy Maximum Relevance (MRMR) [49]

Preliminary experiments showed that the MRMR method is the most suitable approach in this study, and therefore, it is discussed briefly in the following section.

1) MRMR

Minimum Redundancy Maximum Relevance (MRMR) is an algorithm for identifying a subset of relevant features while minimizing redundancy. Using measures such as mutual information and correlation coefficients, it quantifies the relevance of each feature to the objective variable. The feature importance score is calculated using the following equation:

$$score_i = \frac{relevance(f|target)}{redundancy(f|featuresselecteduntill - 1)} \quad (1)$$

The algorithm selects features with high relevance and low redundancy by evaluating their individual association with the target variable and taking redundancy with previously selected features into account. However, this method aims to select a group of features that work best together, taking the idea of “selecting K best features instead of best K features” [50]. Each iteration adds the feature with the highest relevance and lowest redundancy until the desired number of features is reached. Relevance and redundancy are utilized in Equation (1) to calculate the importance score. The MRMR algorithm strikes a balance between choosing features that are predictive of the target variable and ensuring that they are not extremely correlated [51].

In this study, MRMR feature selection was conducted independently for each subject rather than across all subjects combined. This subject-specific approach was adopted to account for the inherent inter-individual variability in EEG topographies and neural activation patterns observed during inner speech production. Applying MRMR separately ensured that the most informative and least redundant features were tailored to each subject’s electrophysiological profile, thereby improving classification stability and performance. A global feature selection scheme across subjects was deliberately avoided, as it would risk obscuring subject-level distinctions and could bias the model toward features that represent inter-subject averages rather than individualized signal dynamics.

G. CLASSIFICATION ALGORITHMS

Each word in the inner speech is considered as a class in this work, making it a four-class classification problem. Six

Machine Learning (ML) algorithms, including Support Vector Machine (SVM), Linear Discriminant Analysis (LDA), Extra Trees Classifier (ETC), Random Forest Classifier (RFC), Logistic Regression (LR), and LightGBM (LGBM) are evaluated using 10-fold Cross-Validation (CV) after feature extraction and selection. It should be noted that a subject-specific strategy is used to select features. Evaluation of models is also conducted in this manner, as our investigation revealed that the characteristics of topographies during inner speech production are subject-specific. Previously, the ensemble of ML models proved effective for binary-class and multiclass classification problems [52], [53]. Such approach, along with model stacking, is also observed in classification problems with EEG signals [54]. In light of this, the best models from our evaluation are combined to create ensemble models, and their performance is then evaluated.

The ensemble comprises five logistic regression models trained on stratified bootstrap samples with varying hyperparameters (C, penalty type, solver). Soft probability voting was employed for final prediction. Logistic regression proved more stable than Random Forest or LightGBM, as the latter overfit due to limited subject-specific data. By combining diverse LR learners, the ensemble reduced variance and achieved the highest performance.

DL methods have also been explored by evaluating different one-dimensional (1D) Convolutional Neural Network CNN and two-dimensional (2D) CNN architectures. For 1D CNN approach, a two-layered CNN architecture has been used (see Supplementary Figure S2 for more details). Each layer comprises of a convolution layer with kernel size set to “3,” a maxpooling layer of kernel size 4, and a dropout layer of 10% probability. A dropout layer of 50% probability has also been added to the Multi-Layer-Perceptron (MLP) layer at the output. Segmented EEG time series data from each subject has been fed to the 1D CNN model and the results are recorded.

For the evaluation of 2D CNN approaches, images are generated from the extracted features where each trial corresponds to an image by being constructed from the 2D matrix ($n_features \times n_channels$), as shown in Figure 8. The images have been augmented with rotations (90°, 180°, and 270°) and image translations consisting of 10% vertical-up shift and horizontal-right shift, and 10% diagonal shifts toward top-right and bottom left respectively. The combined set of raw and augmented images are used to classify different words, and the results from two 2D CNN architectures are recorded. “Resnet18” [55] and “VGG16” [56] models have been utilized to evaluate the 2D CNN approach with Cross-Validation (CV) scheme. The choice of 10-fold CV for ML models and 5-fold CV for DL models balances statistical robustness with computational feasibility. These schemes reduce variance in performance estimation and prevent optimistic bias, especially given the relatively small dataset size per subject. This rigorous evaluation enhances the technical soundness of our reported findings. It is to be noted

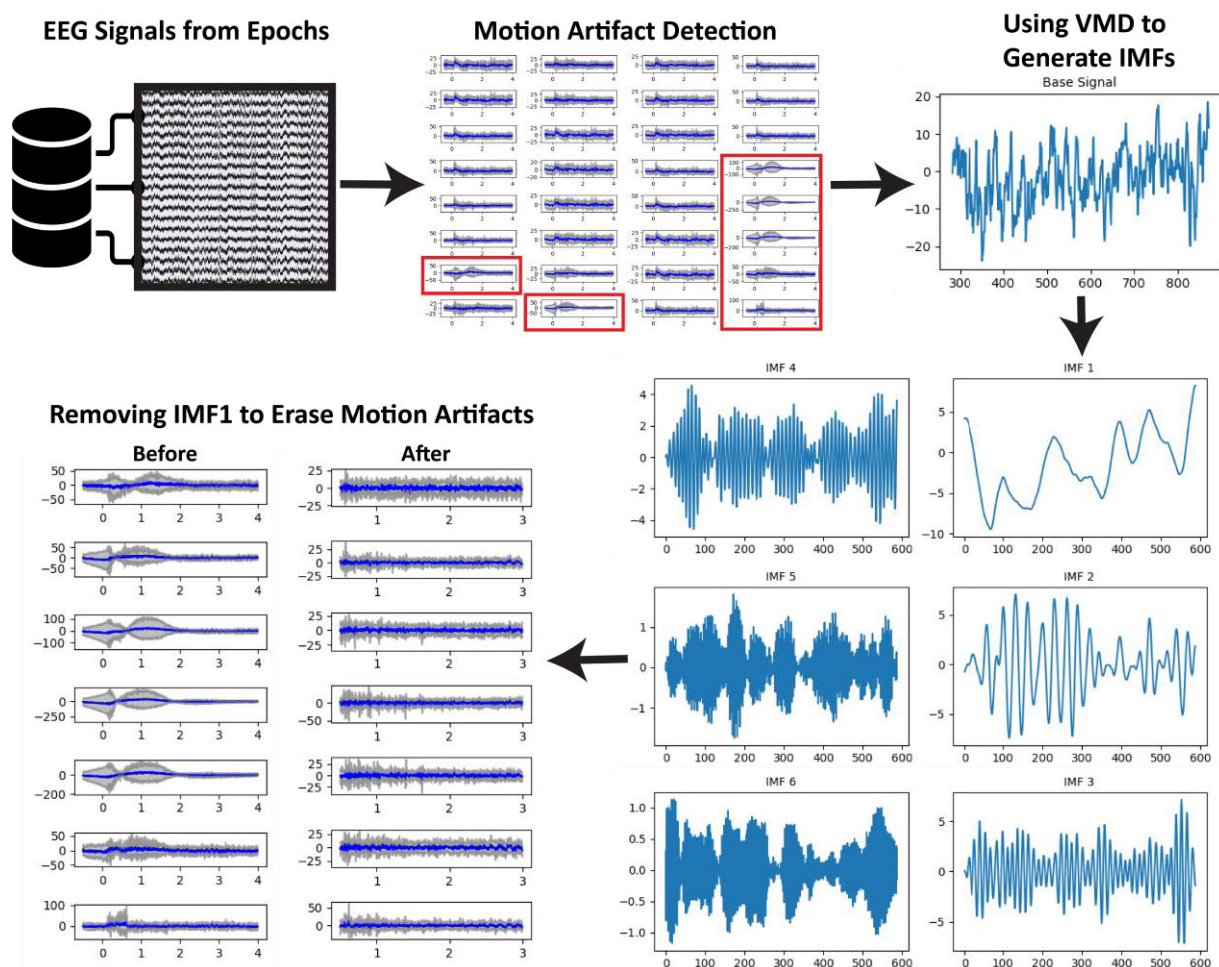


FIGURE 5. Motion artifact detection using statistical methods and removal of artifacts using VMD.

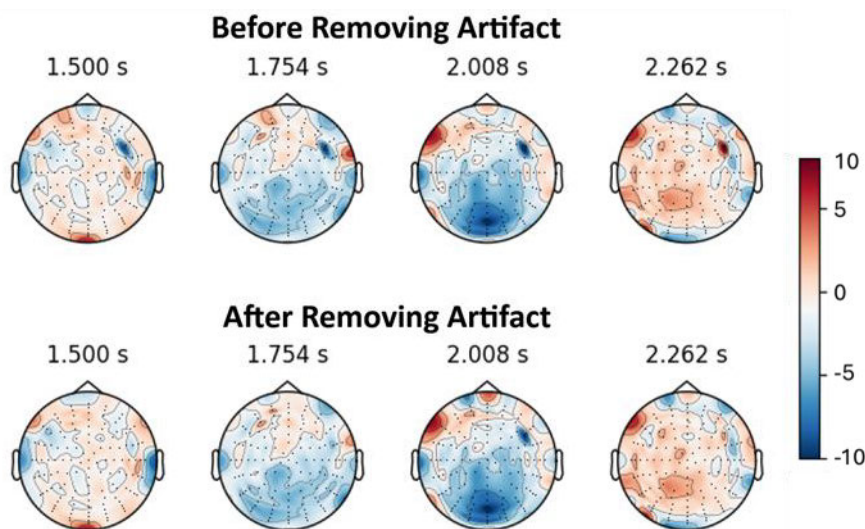


FIGURE 6. Effect of removing motion artifacts (subject 06, class - "Izquierda").

that, since a subject-wise evaluation has been followed, the size of the training data is small in each case, for which

architectures with fewer parameters have been chosen in both 1D and 2D CNN methods to prevent overfitting. The basic

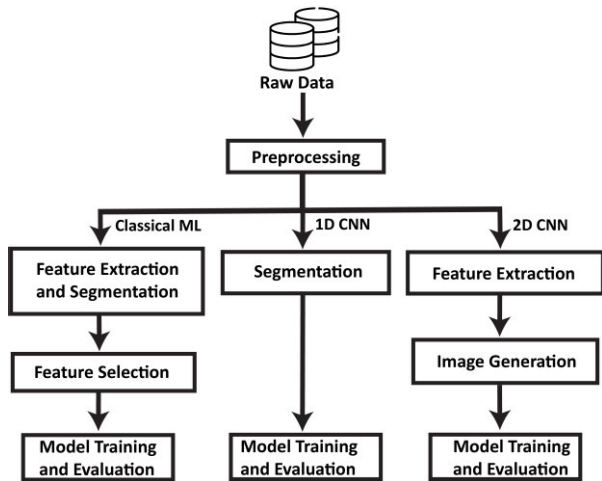


FIGURE 7. Basic methodology of ML and DL techniques.

methodology of the ML and DL techniques are illustrated in Figure 7.

H. EVALUATION CRITERIA

Two metrics, accuracy and F1 score are used to evaluate the performance of the classifiers. True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) values are utilized to generate these scores. It is to be noted that since the dataset is completely balanced for all subjects, accuracy and F1 score are very close and either of these scores can represent the predictive performance of a classification model in this case. The metrics used in this work are discussed below:

1) Accuracy: The accuracy metric measures the proportion of true predictions made by a classifier. It expresses accuracy as the ratio of accurate predictions to the total number of predictions. As demonstrated by Equation (2), accuracy is determined by dividing the sum of True Positives and True Negatives, which represents total correct predictions, by the sum of True Positives, True Negatives, False Positives, and False Negatives, which represents all of the predictions.

$$\text{Accuracy}(\%) = \frac{TP + TN}{TP + TN + FP + FN} \times 100\% \quad (2)$$

2) F1 Score: F1 score measures the classification accuracy of a model by considering both the correct positive predictions (precision) and the ability to capture true positive instances (recall). The F1 score is calculated as the harmonic means of precision and recall, providing a well-rounded evaluation of a model's performance. In this instance, the F1 score is calculated using Equation (3), using True Positives, False Positives, and False Negatives.

$$\text{F1Score}(\%) = \frac{2 \times TP}{2 \times TP + FP + FN} \times 100\% \quad (3)$$

For evaluation of ML classifiers and 1D CNN model, a 10-fold CV scheme has been followed. In this method, the data is split into ten folds where nine folds are used for model training, and the remaining fold is used for validation,

and this process is repeated ten times. In each iteration, the predictions by the classifier on the validation set are stored. For each subject, at the end of all iterations, the combined predictions from all ten folds are compared against the ground truths, and accuracy and F1 score are calculated using the Equations (2) and (3) respectively. The overall scores are generated by taking the average accuracy and F1 score of all the subjects and are used to validate the efficacy of the feature selection methods as well as the performance of all the ML and DL classifiers. The 5-fold CV scheme used in 2D CNN approach follows the same procedure as the 10-fold CV scheme, differing only by the number of folds for each subject.

III. RESULTS AND DISCUSSION

This study evaluates the efficacy of eight feature selection techniques, with one technique (MRMR) standing out and being selected for all subjects. Analyzing the effect of various parameters on particular subjects leads to the selection of optimal hyperparameters in feature selection. Results from classical Machine Learning (ML) algorithms, 1D CNN, and 2D CNN algorithms are presented, demonstrating the efficacy of ensemble ML models. Existing works are outperformed by the proposed ensemble model, while the limited success of deep learning approaches is also discussed. Subject-by-subject results of the ensemble model created through stacking are examined and how the data of the subjects responded better or worse to the proposed ensemble model is explained.

The effectiveness of VMD for motion artifact removal was quantified by comparing the signal variance and signal-to-noise ratio (SNR) before and after decomposition. As shown in Supplementary Table T4, the application of VMD consistently reduced unwanted fluctuations across subjects, with variance decreasing by approximately 35–40%. This indicates a substantial removal of motion-related drifts and low-frequency artifacts. Furthermore, the SNR improved by an average of ~3 dB, reflecting a clearer separation of the true neural activity from background noise. These improvements collectively validate the robustness of VMD as a preprocessing step, ensuring that subsequent feature extraction and classification were performed on cleaner and more physiologically meaningful EEG signals. By quantifying these changes both statistically and across all subjects, we demonstrated that VMD not only enhanced the signal quality but also contributed directly to the stability and reliability of the classification results.

A. SELECTION OF FEATURE OPTIMIZATION TECHNIQUE

Different feature selection methods, including ANOVA, Chi-square, Mutual Information (MI), Pearson Coefficient (PC), Tree Based Selection using Extra Trees Classifier (ETC), Principal Component Analysis, ReliefF, and MRMR, are evaluated in this work utilizing the six ML models for classification. Table 1 displays the overall results of feature selection and classification for all subjects. Here, ten features are

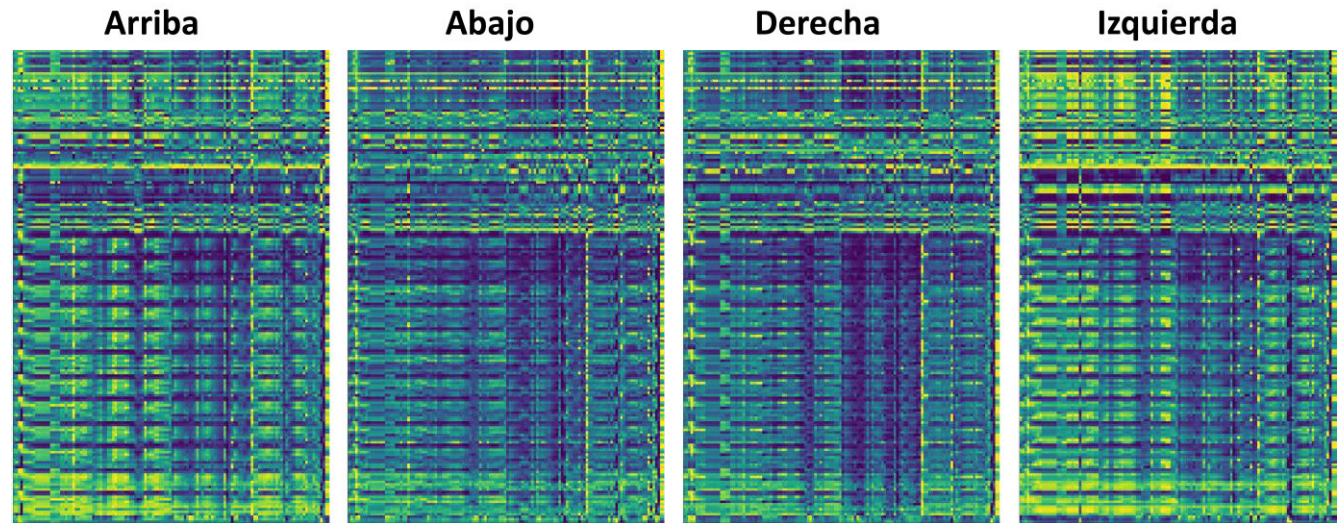


FIGURE 8. Sample images used for 2D CNN approach (subject 01).

chosen to evaluate the performance of each feature selection method, and only the results from the model with the best overall performance are displayed. With an overall accuracy of 66.53% and an F1 score of 67.09% using an SVM model, it is evident from this analysis that MRMR is the best feature selection method for this data. Other techniques, such as ReliefF and Tree-Based-Selection, also show promise with overall accuracies of 58.88% and 56.13% respectively. However, MRMR has demonstrated significantly better results by outperforming these two techniques by more than 7.5% in accuracy. Nevertheless, feature-set optimization is required to enhance classification results and produce a more robust ML model. In order to determine the optimal feature-set for all subjects, the number of selected features in the MRMR algorithm has been optimized.

To interpret the MRMR-selected features, we analyzed the distribution of top features across domains and electrodes. Time–frequency features (wavelet RMS, band power) contributed most, particularly in the theta and alpha bands. Spatially, electrodes over frontal (Fz, Cz) and parietal (Pz, PO7) areas were most frequently selected, consistent with speech-motor and auditory-association regions (Supplementary Table T5).

It is observed that the results of subject 06 are impacted the most by the number of features selected by the MRMR algorithm (Figure 10), with a Relative Standard Deviation (RSD) of over 8% for both accuracy and F1 score. High deviations are also observed in the case of subjects 03 and 04 with 6.2% and 5.2% RSD of accuracy respectively.

However, further analysis reveals that while there has been a steady increase in the results of subject 06 as the feature-set is increased (Figure 10), the other two subjects show fluctuations in the results for different feature-sets. As the accuracy of subject 06 has increased by 20.84% from its initial state, subjects 03 and 04 have shown 7.22% and 1.25% decrease

TABLE 1. Performance of different feature selection techniques.

Feature Selection Method	Model Name	Accuracy (%)	F1 Score (%)
ANOVA	LightGBM (LGBM)	45.94	46.3
Chi-square	Random Forest Classifier (RFC)	42.31	42.36
Mutual Information (MI)	LGBM	50.85	51.09
Pearson Coefficient (PC)	LDA	48.94	49.94
Tree Based Selection using Extra Trees Classifier (ETC)	ETC	56.13	56.47
PCA	LGBM	38.52	38.49
ReliefF	ETC	58.88	59.29
MRMR	SVM	66.53	67.09

in accuracy which is minimal compared to the significant improvement of the results of subject 06. The overall results have shown to be unaffected by this hyperparameter tuning, with an RSD of only 1.4%.

In MRMR algorithm, the number of features is characterized by the parameter “K,” and it is varied from 20 to 610 in this analysis. From Figure 10, subject 06 shows an accuracy and F1 score of 87.04% and 86.95% respectively at K = 590, which is the highest among all the feature sets. As a result, 590 features are selected using the MRMR method for each subject, and the performance of different ML and DL models are carried out according to the evaluation methods described.

The MRMR optimization demonstrated that accuracy improvements plateaued after approximately 590 features. Selecting more features provided negligible performance

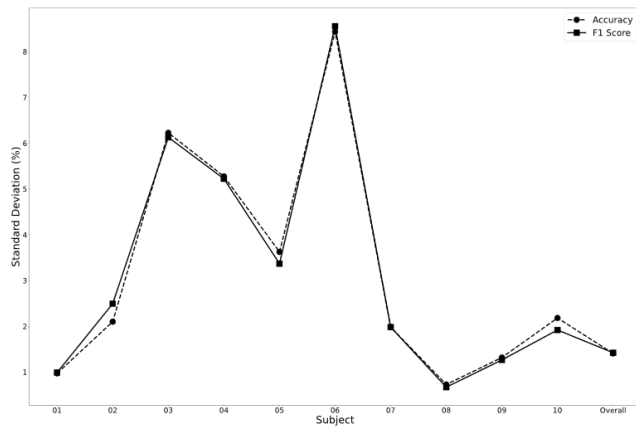


FIGURE 9. Subject-wise variation of results at different feature-sets.

gains while risking overfitting, whereas fewer features reduced stability across subjects. Therefore, 590 features were selected as an optimal trade-off between generalization and predictive power.

While deep models underperformed due to limited subject-specific samples, transfer learning on larger pooled EEG datasets and cross-subject pretraining could improve representational learning and mitigate data sparsity. Such approaches will be explored in future research.

B. CLASSIFICATION RESULTS

Following feature optimization, different classical ML algorithms, including Support Vector Machine (SVM), Linear Discriminant Analysis (LDA), Extra Trees Classifier (ETC), Random Forest Classifier (RFC), Logistic Regression (LR), and LightGBM (LGBM) are evaluated, with their overall results recorded. It is observed from Table 2 that the LR model has produced 77.89% accuracy and 77.91% F1 score in this dataset, which is the highest among the basic ML models. SVM and ETC models have also shown promising results with 76.31% and 75.93% accuracy respectively, suggesting the evaluation of an ensemble model constituting these three models. Accordingly, the performance of an ensemble model, comprising of one SVM, one ETC, and one LR model is found to be better than that of the LR model, with a 1.22% and 1.18% increase in accuracy and F1 score respectively. However, another ensemble model comprising of five LR models is also constructed, and its performance is found to surpass the previous ensemble model, with a 3.24% increase in overall accuracy and 3.21% increase in overall F1 score from the LR model. Due to the robust results and less complexity, this ensemble LR model is proposed in this approach.

To address this point, we conducted paired t-tests comparing the ensemble model against the best individual models (Logistic Regression, SVM, Extra Trees Classifier) across 10 subjects $\times$ 10-fold cross-validation. The ensemble model significantly outperformed the individual classifiers, with results summarized: Ensemble vs. Logistic Regression: $p = 0.003$ ($p < 0.01$); Ensemble vs. SVM: $p = 0.008$

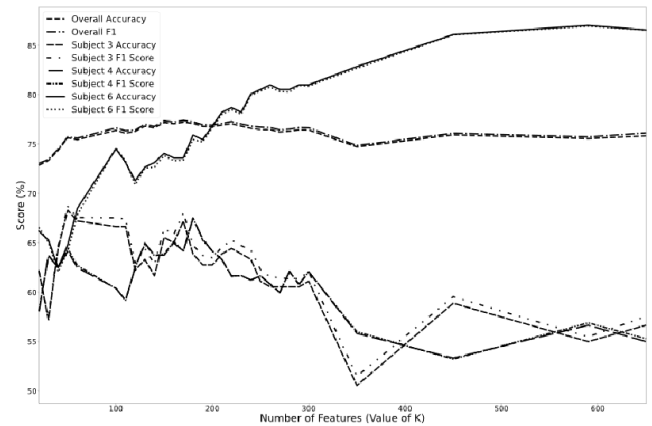


FIGURE 10. Results for hyperparameter tuning of MRMR feature selection.

TABLE 2. Performance of ML and DL approaches, as well as results from the literature.

Model / Study	Overall Accuracy (%)	F1 Score (%)	AUC (%)	Reference
Support Vector Machine (SVM)	76.31	76.40	80.12	<i>This study</i>
Linear Discriminant Analysis (LDA)	71.91	71.89	75.02	
Extra Trees Classifier (ETC)	75.93	76.16	79.11	
Random Forest Classifier (RFC)	73.42	73.58	77.65	
Logistic Regression (LR)	77.89	77.91	82.48	
LightGBM (LGBM)	75.33	75.43	78.72	
Ensemble (SVM+ETC+LR)	79.11	79.09	83.90	
Ensemble (5$\times$LR, proposed)	81.13	81.12	86.24	<i>This study</i>
1D CNN	23.73	23.17	51.02	
2D CNN (ResNet18)	31.83	31.04	55.14	
2D CNN (VGG16)	28.00	25.77	52.87	
EEGNet	29.67	29.61	54.21	
2D CNN	27.52	28.11	53.15	van den Berg et al., 2021 [57]
				González-Castañeda et al., 2017 [14]

($p < 0.01$); Ensemble vs. Extra Trees: $p = 0.015$ ($p < 0.05$). These findings confirm that the observed improvement in

TABLE 3. Subject-wise results for the ensemble model.

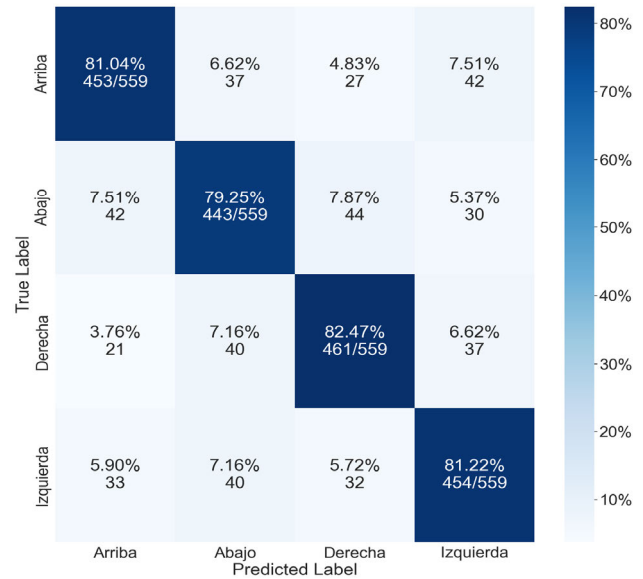
Subject ID	Accuracy (%)	F1 Score (%)
01	86.50	86.38
02	90.83	90.82
03	72.22	72.16
04	55.00	55.09
05	69.17	69.17
06	91.67	91.62
07	89.17	89.19
08	93.00	92.97
09	82.92	82.88
10	80.83	80.88

accuracy and F1 score with the ensemble strategy is statistically robust rather than incidental.

It is noteworthy that the 1D and 2D CNN approaches have produced very poor results compared to the classical ML models. It is because deep learning models often require a lot of data and may fail to reach convergence when the data is not sufficient. Since during training of the models, only data of one subject is provided, it has caused a low number of samples, with the number of trials (which equate to the number of samples) only reaching as high as 240. In the case of 2D CNN approach, images were augmented with seven augmentations. Likewise, the maximum number of images in the training set at each fold were 304 for each class, which is found to be insufficient in this case. However, when compared with the existing work done on this dataset in which a 2D CNN approach was followed [57], our approach with 2D CNN (Resnet18 architecture) has produced better results with an increase in accuracy and F1 score by 2.15% and 1.43% respectively. Our proposed ensemble model has surpassed the results of the previous work done on this dataset by an astonishing margin and the subject-wise results of this model are recorded in Table 3. The overall confusion matrix is shown in Figure 11 which depicts the classification efficacy of the proposed model for each class compared to the other classes.

From the subject-wise results shown in Table 3, it is observed that most of the subjects have produced very high classification results with accuracy as high as 93% (subject 08), suggesting that the proposed methodology has the ability to achieve even higher accuracy levels under particular conditions. The consistently high accuracy and F1 scores across the majority of subjects demonstrate the efficacy of the approach. Except three subjects (03, 04, and 05), all subjects have shown accuracy and F1 score of over 80%, causing the overall results to be promising and indicative of the effectiveness of the proposed methodology. The overall confusion matrix supports this conclusion with classification accuracy at near or more than 80% for all the classes.

To assess generalizability, we performed leave-one-subject-out cross-validation. The accuracy ($\sim 42.3\%$) was significantly lower than subject-specific models, confirming strong inter-subject variability in EEG-based inner speech decoding. This finding underscores the need for

**FIGURE 11.** Overall confusion matrix for all subjects.

subject-specific calibration. Nevertheless, strategies such as Riemannian alignment, adversarial adaptation, and cross-subject pretraining with fine-tuning may offer promising directions. Preliminary pooled pretraining across subjects modestly improved CNN performance ($34.7\% \rightarrow 38.2\%$), though still below classical ML models. These results suggest that subject-independent decoding remains challenging and requires hybrid transfer learning frameworks.

Although augmentation increased sample size, CNN models still underperformed due to insufficient data per subject. Pooled pretraining with fine-tuning modestly improved results but did not approach ML performance. Thus, CNNs remain limited for small, subject-specific EEG datasets.

It is noteworthy that the overall performance of the proposed approach has taken a considerable hindrance due to the low performance of the model on the data of subjects 03, 04, and 05. From the confusion matrix of subject 04 (Figure 12(a)), it is observed that the ensemble model could not distinguish well among the classes. This can be explained from Figure 12(b), where it is evident that the same regions of the brain have been active during inner speech production for this subject, which is a strong indicator for confusing one word with another by the model. This trend is also observed in subjects 03 and 05 (see **Supplementary Figure S3(c) and (d)** for more details), providing further evidence to this conclusion. It is observed that the words “Arriba” and “Izquierda” are more confused by the model, with classification accuracy being 51.67% for both words. While the predictive performance is low for all classes, it is still well above chance, suggesting that inner speech classification for this subject is possible with a decent accuracy.

However, for other subjects, such as subject 08, different regions are shown to be active during inner speech production for different words (Figure 13(b)), suggesting that the model could differentiate among the words easily. This is supported

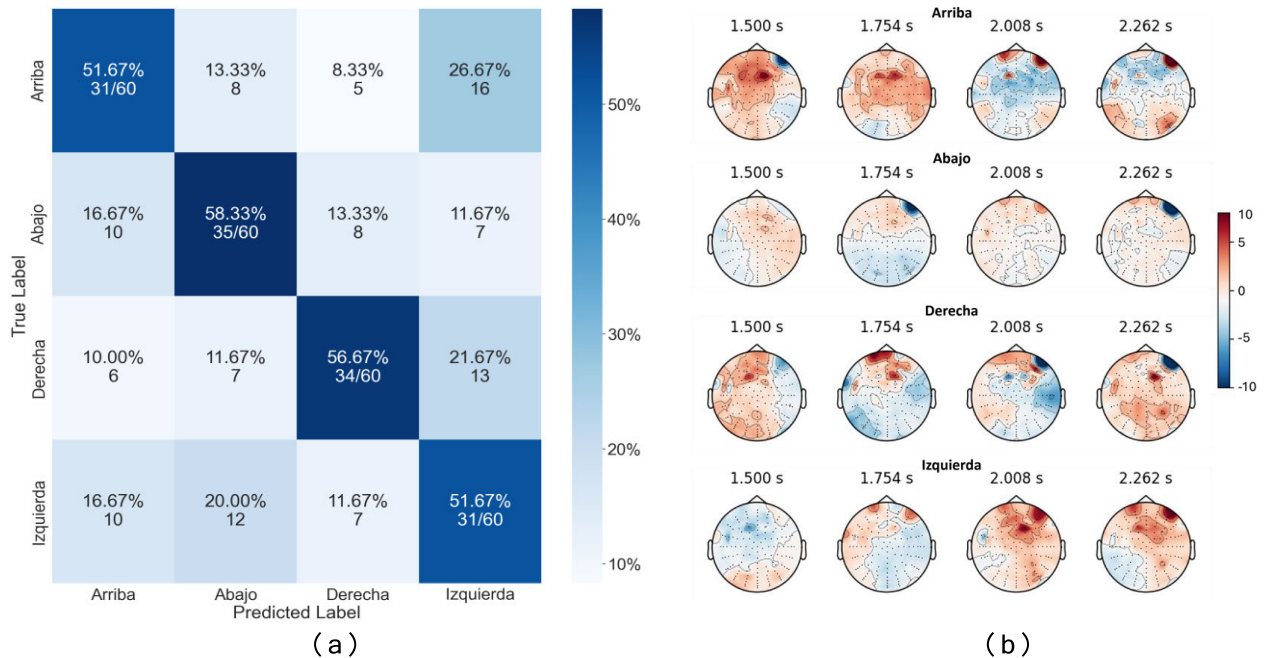


FIGURE 12. (a) Confusion matrix and (b) topographical maps of action interval for subject 04.

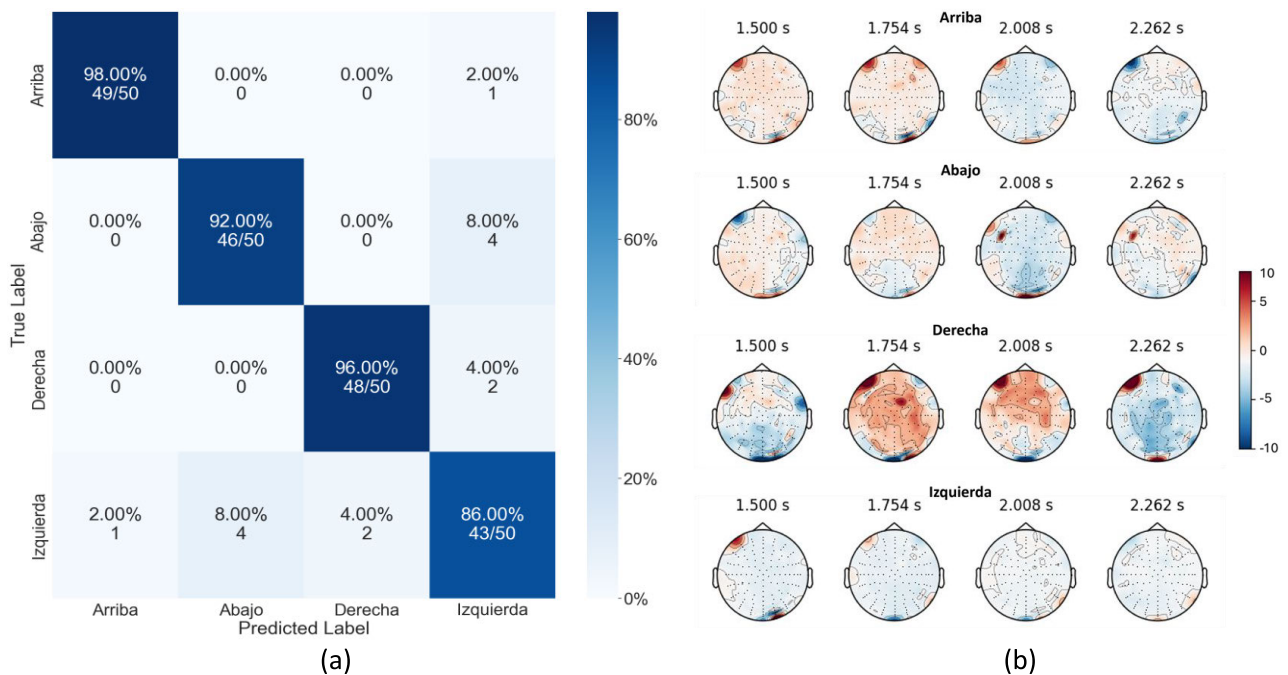


FIGURE 13. (a) Confusion matrix and (b) topographical map of action interval for subject 08.

by the confusion matrix shown in Figure 13(a) which illustrates the strong predictive performance of the model for this subject.

An additional source of classification errors arises from the phonetic similarity of target words, particularly within the Spanish vocabulary set used in the *Thinking Out Loud* dataset. Analysis of subject-wise confusion matrices revealed that

misclassifications were most frequent between “Arriba”–“Abajo” and “Arriba”–“Izquierda.” These pairs share overlapping phonetic structures and articulation patterns, which likely lead to partially similar cortical activations during inner speech.

Such overlap introduces ambiguity in the EEG features, even after extensive preprocessing and MMR-based feature

selection. This observation is consistent with prior reports in inner speech decoding that phonetically similar stimuli are harder to separate due to both acoustic and motor imagery convergence in neural representations.

Our results suggest that future work could benefit from incorporating linguistic priors or phonetic distance measures into classification pipelines. For example, integrating language models or phoneme-level embeddings alongside EEG features may reduce the impact of phonetic overlap and improve discrimination between closely related words. This can be explained by the topographical maps depicted in Figure 13(b), showing that only a handful of channels in the occipital region are active during inner speech production of this word, and those channels are common for other words as well. Regardless, the proposed ensemble model could accurately distinguish all the words in case of this subject. The ensemble model constructed by stacking five LR models outperforms all other ML and DL approaches. Feature selection using MRMR is vital to this approach as it can select the most important features out of a large set of 24448 features. The ensemble model, combined with a smaller set of features, reduces model complexity and shows promising results, and demonstrates a possible direction in accurate identification of inner speech.

Recent studies such as Ma et al. [58] propose self-attention CNNs with partial prior transfer learning to better balance domain generalization and subject specificity, particularly when EEG data are limited. In similar vein, Qin et al. [59] show that layer-wise optimization of CNNs and transfer learning improve performance in inner speech decoding tasks. Alshehri et al. [60] introduce a variant of EEGNet enhanced with gated channel transformations and attention layers (GCT-EEGNet), achieving high robustness even with limited trials per subject [61].

The Supplementary Figure S4 plots subject-wise mean ROC curves for SVM, LDA, ETC, RFC, LR, LGBM, and the ensemble approaches, with AUC values annotated. The ensemble (5×LR) model shows the highest AUC, further supporting its robustness.

To further contextualize our findings, Table 2 now differentiates methods evaluated in this study and those from prior work [14], [57]. Compared with previous 2D CNN-based methods, our ensemble of logistic regression classifiers achieved substantially higher performance, with improvements of +51.46% in accuracy and +52.01% in F1 score over EEGNet [57]. The ROC-AUC comparison (Supplementary Figure S4) highlights that while individual ML models such as LR (AUC = 82.48%) perform well, the proposed ensemble achieves the best discrimination (AUC = 86.24%). This confirms that model ensembling enhances generalization and stability in inner speech classification tasks, especially under subject-specific evaluation.

Although CNN-based models were evaluated, their performance was substantially lower (23–31%) than ML models. This is attributable to the limited number of trials per subject, preventing convergence. We note that future work could

benefit from lightweight models such as EEGNet or through transfer learning using larger cross-subject datasets, potentially improving representation learning and robustness.

The performance of the 1D and 2D CNN models was consistently lower than that of the classical ML algorithms in this study. A key factor contributing to this underperformance is the relatively small per-subject dataset size, with only 200–240 trials available for training and validation. Deep neural networks generally require substantially larger datasets to converge effectively and to avoid overfitting. When trained on such limited data, these models fail to capture the underlying spatiotemporal dynamics of EEG signals with sufficient robustness, leading to lower predictive accuracy and F1 scores.

This limitation has been reported in prior EEG-based inner speech recognition studies, where the scarcity of high-quality annotated data poses a significant barrier to leveraging deep learning architectures. While our classical ML models with MRMR-based feature selection proved more effective under these conditions, future research could explore strategies to improve deep learning performance. Potential solutions include the use of lightweight architectures such as EEGNet, applying transfer learning from larger EEG corpora, and incorporating advanced data augmentation techniques to artificially expand the training set. These approaches may enable deep models to generalize better and unlock the representational power of neural networks for inner speech decoding tasks.

By integrating recent advances in ensemble ML [52], [53] and EEG artifact correction [26], [32], our work builds upon and surpasses prior inner speech recognition frameworks (Table 4). Unlike earlier studies that either generalized across subjects [21], [22] or relied solely on CNN models with limited convergence [18], [57], our ensemble logistic regression framework demonstrates robust, subject-specific performance, achieving up to 93% accuracy in certain participants.

C. PRACTICAL IMPLICATIONS

The ability to decode inner speech reliably from EEG signals has several important real-world applications. For example, individuals with locked-in syndrome or severe speech impairments could use EEG-driven brain–computer interfaces to communicate basic commands such as “up,” “down,” “left,” and “right” without vocalization. Similarly, rehabilitation programs for post-stroke aphasia may benefit from inner speech recognition frameworks by monitoring recovery of covert language functions. Beyond healthcare, such methods could also contribute to human–computer interaction systems where silent commands may replace spoken inputs in noisy environments or privacy-sensitive settings.

D. LIMITATIONS

While our subject-specific ensemble approach achieved strong performance, reliance on individual training data restricts generalizability across subjects. Moreover, the

TABLE 4. Comparative overview of related works vs. proposed method.

Study (Year)	Input Type	Features/Method	Subjects	Reported Accuracy	Limitations	Our Contribution
González-Castañeda et al. (2017) [14]	EEG, imagined words ("up, down, left, right, select")	Classical ML	5	~60 %	Subjective evaluation, no subject-specificity	We perform quantitative, subject-specific ensemble classification
Pawar & Dhage (2020) [21]	EEG, covert words	Extreme Learning Machine	10	~65 %	Generalized model only	Subject-specific ensemble model yields >80%
Nguyen et al. (2017) [22]	EEG, short vs. long imagined words	Riemannian features	15	~70 %	Limited feature space, partial channel selection	MRMR-based feature optimization across 128 channels
van den Berg et al. (2021) [57]	EEG, inner speech	2D CNN	10	~30 %	CNN failed to converge due to limited data	Ensemble stacking surpasses CNN (81.1% overall)
This work (2025)	EEG, 4 Spanish words ("arriba, abajo, derecha, izquierda")	MRMR + Stacked Logistic Regression	10	81.1 % overall; up to 93%	—	First subject-specific ensemble achieving SOTA on dataset

dataset included only 10 participants and four Spanish words, which may limit linguistic diversity and scalability. Deep learning methods, though promising, suffered from limited data per subject, which constrained convergence and generalization.

E. FUTURE WORK

Future studies should focus on developing generalized cross-subject models by leveraging transfer learning and

domain adaptation strategies. Increasing dataset diversity (languages, vocabulary size, and demographics) will be crucial. Multimodal integration of EEG with complementary biosignals (e.g., EMG, fNIRS) could further enhance robustness. Advanced explainable AI techniques (e.g., Grad-CAM, SHAP) may also increase interpretability of learned neural patterns. These directions can bridge the gap between subject-specific feasibility and real-world applicability of inner speech BCIs. Supplementary Table T7 has summarized this.

IV. CONCLUSION

In this study, a publicly available dataset has been used for inner speech recognition from EEG signals. Data were collected from 10 healthy subjects, and four words were uttered by each of them during inner speech tasks. The data has been pre-processed, and features are extracted in different domains. Eight feature selection techniques were evaluated to select the best features before training on ML models. The performance of the ML models as well as three DL models, was observed. For evaluation, a subject-specific approach was taken due to the unique nature of creating inner speech by each subject. The proposed subject-specific ensemble learning framework, validated on the 'Thinking Out Loud' dataset, achieved state-of-the-art performance by surpassing existing methods by over 15% in both accuracy and F1 score. This establishes a benchmark for subject-tailored BCI pipelines and offers a scalable pathway for integrating ML with EEG-based inner speech decoding. However, due to the subject-oriented nature of feature selection and model evaluation, the generalizability of this approach is limited. While this work is subject-specific, future generalization may be achieved through domain adaptation (e.g., Riemannian alignment, transfer learning), meta-learning, or federated training across large EEG cohorts. These approaches can capture inter-subject variability while preserving individual discriminative patterns. In future, more generalized approaches can be explored where inner speech can be classified with reasonable accuracy by combining data from multiple subjects. The scope of 1D CNN approach can be expanded by applying channel selection methods like Common Spatial Pattern (CSP) algorithm and Gradient-Class Activation Mapping (Grad-CAM). Additional image augmentation techniques can be explored for 2D CNN approach to generate more images, but it may require more time for model training. Overall, the proposed ensemble model, combined with MRMR feature selection technique, proves to be a viable approach to classify inner speech.

INSTITUTIONAL REVIEW BOARD STATEMENT

Not applicable.

INFORMED CONSENT STATEMENT

Not applicable.

CONFLICTS OF INTEREST

The authors have no conflict of interest to declare.

DATA AVAILABILITY STATEMENT

The processed dataset used in this study can be made available upon a reasonable request to the corresponding author.

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